Children's Hospital LOS ANGELES

USC Viterbi School of Engineering

Cross-scanner translation of bSTAR sequence

ISMRM Virtual Meeting on Pulseq
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Disclaimer

- The technical components of this presentation are largely based on the following publications.
  - Lee NG, Bauman G, Bieri O, Nayak KS. Replication of the bSTAR sequence and open-source implementation. Magn Reson Med. 2024 Apr;91(4):1464-1477.
  - Bauman G, Lee NG, Tian Y, Bieri O, Nayak KS. Submillimeter lung MRI at 0.55 T using balanced steady-state free precession with half-radial dual-echo readout (bSTAR). Magn Reson Med. 2023;90:1949-1957. doi:10.1002/MRM.29757
- Some MRI safety implementation considerations were informed by empirical experience and may not be directly transferable to other MRI systems. Users are strongly suggested to collaborate with local engineers and vendors to ensure compatibility and safety within their specific scanner environment.

Outline

- Motivation
- Sequence design
- Scanner translation
- Results

Motivation

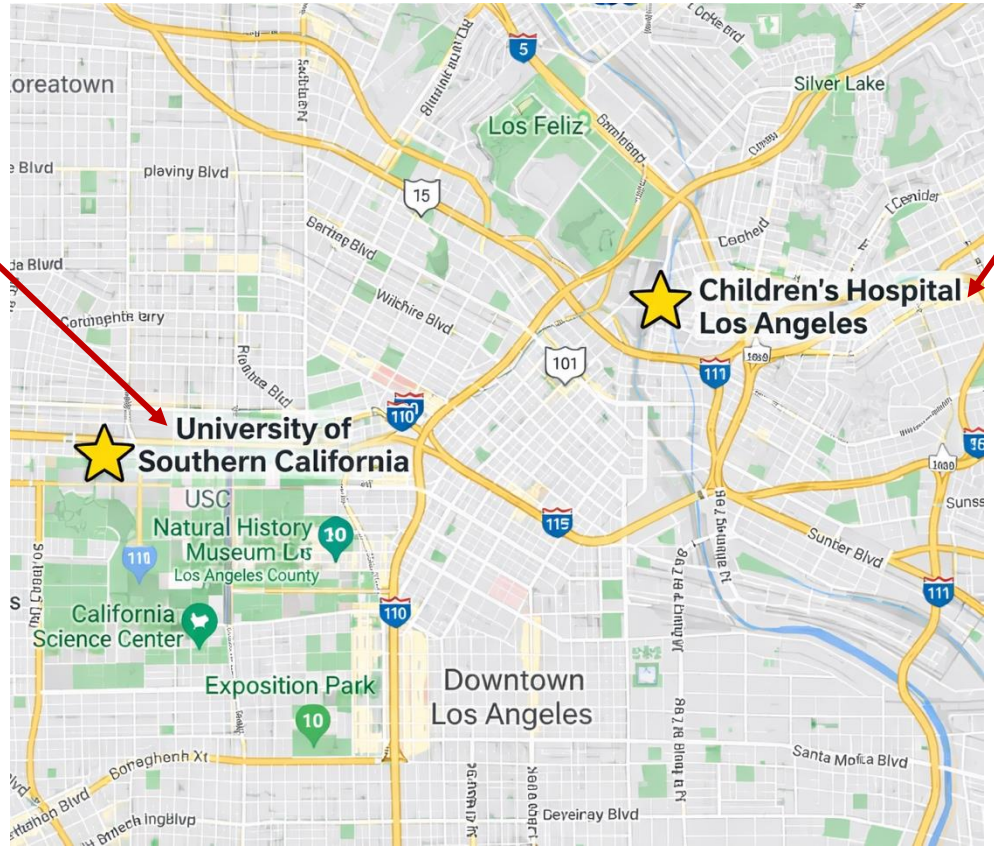
- bSTAR: balanced SSFP sequence with half-radial dual-echo readout. The sequence has shown particularly promising performance in lung imaging with extremely short TRs and high parenchyma SNR. Example: 3D 0.9-mm isotropic resolution lung MRI with bSTAR at 0.55T[1]

[1]Bauman et al. Submillimeter lung MRI at 0.55 T using balanced steady-state free precession with half-radial dual-echo readout (bSTAR). MRM 2023

Motivation

- A 0.55T MRI translational pipeline for pediatric lung imaging.

Prototype system
"ramped-down" Aera
SW: VE11S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Exposition Park
Downtown Los Angeles
Santa Monica Blvd
Sunset Blvd
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Exposition Blvd
Coliseum St
Vernon Ave
Slauson Ave
Florence Ave
Manchester Ave
Imperial Hwy
Firestone Blvd
Rosecrans Ave
Alondra Blvd
Artesia Blvd
Gardena Blvd
Compton Blvd
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Artesia Blvd
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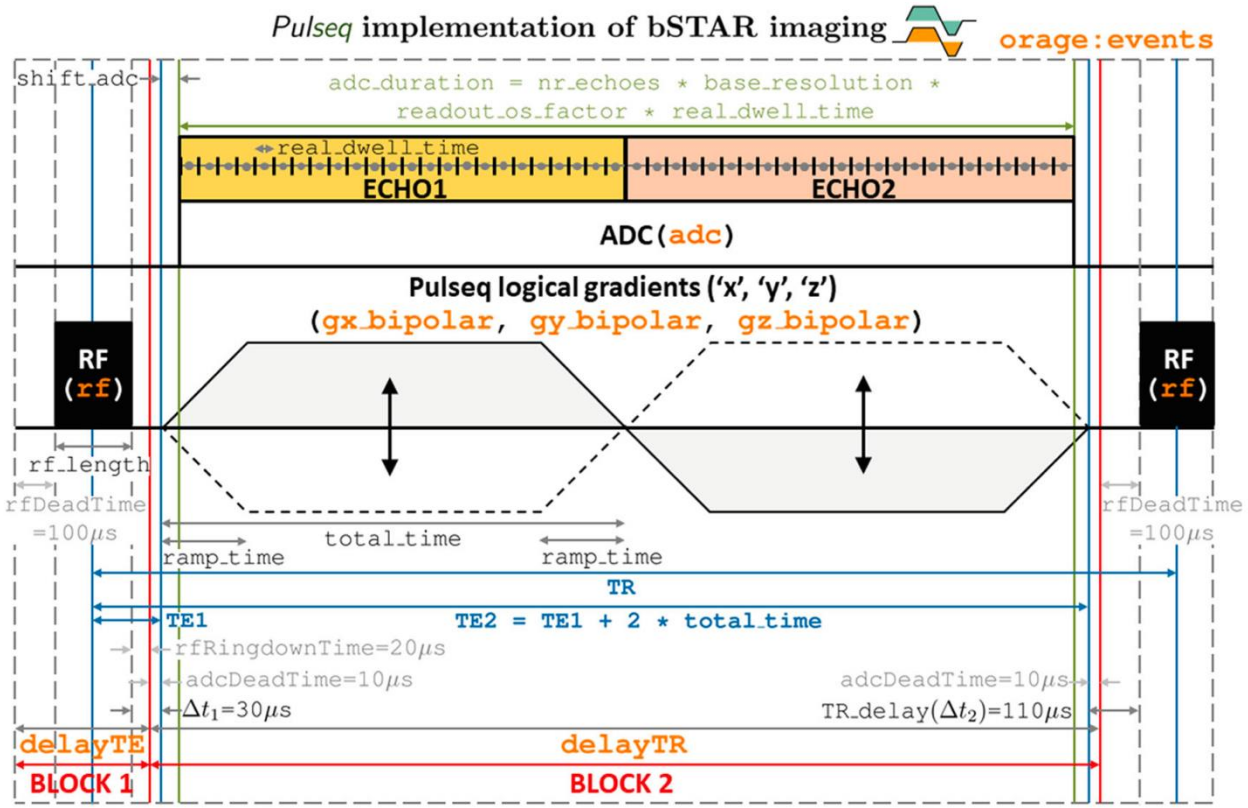
Venice Blvd
Washington Blvd
Jefferson Blvd
Exposition Blvd
Coliseum St
Vernon Ave
Slauson Ave
Florence Ave
Manchester Ave
Imperial Hwy
Firestone

Sequence design

- Components: a non-selective hard RF pulse, one bipolar gradient on each gradient axis, one analog-to-digital converter (ADC) window, and hardware time delays.



Dr. Nam Lee

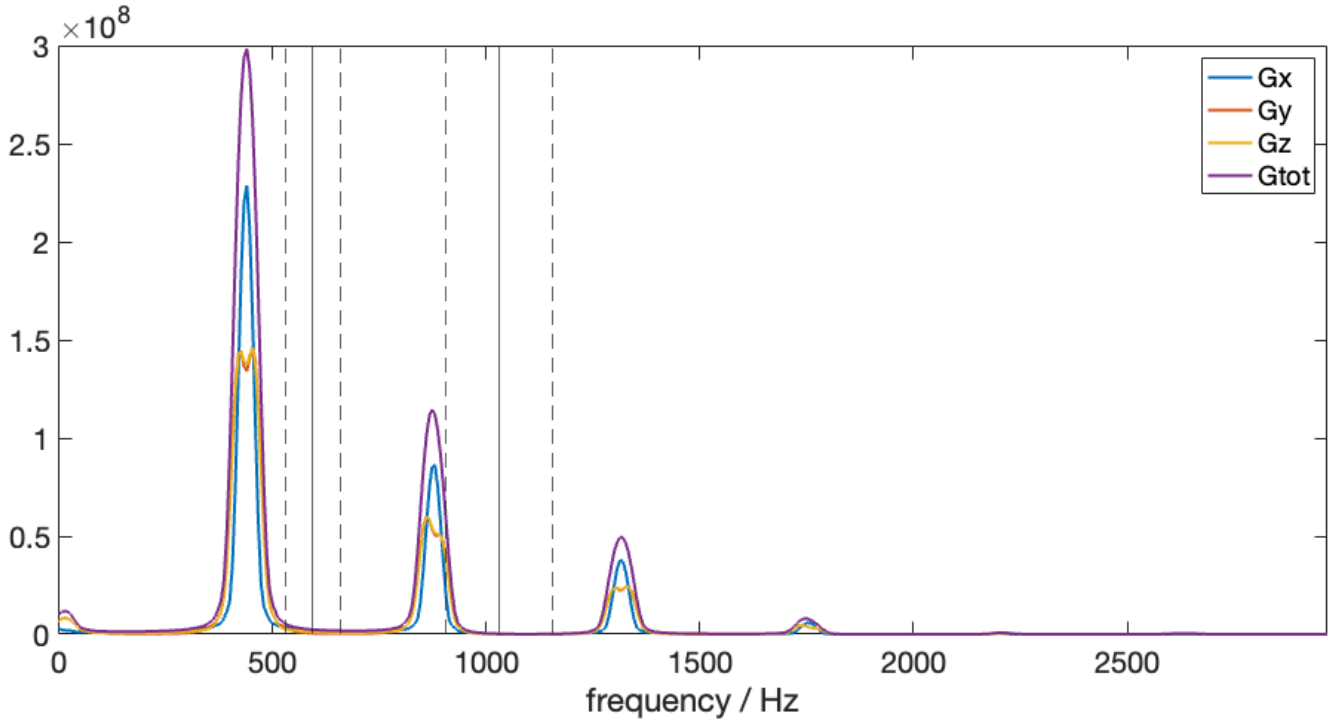
Lee et al. Replication of the bSTAR sequence and open-source implementation. MRM 2024

Pulseq source codes: https://github.com/usc-mrel/replication_bstar

Scanner translation

- **Same codebase**, despite different software and hardware configurations.
- Gradient coil heating: vendor provided gradient temperature logging tool
  - E.g. PerCoilTemp.log from Siemens
- Gradient acoustic resonance frequency forbidden bands:

```
calculate_gradient_spectrum(sys, seq, aflAcousticResonanceFrequency,
                                      aflAcousticResonanceBandwidth, output_path)
```

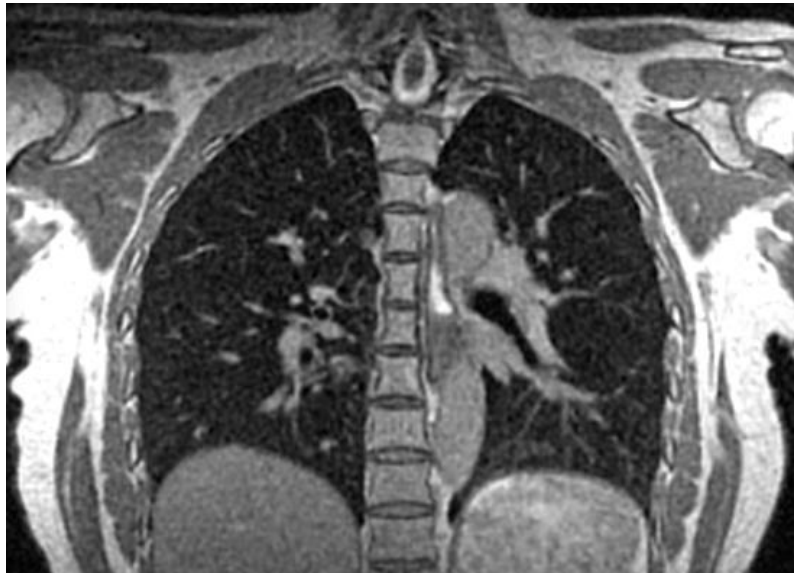


Results

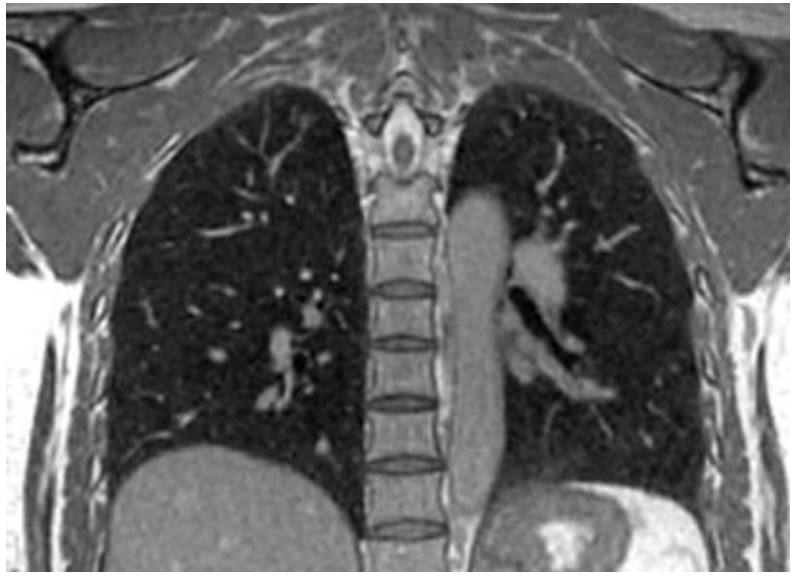
- Cross-scanner validation

Prototype system
0.9mm resolution

Commercial system
1.3mm resolution



Same .seq file
but run @prototype system
1.3mm resolution



Summary

- Lee NG, Bauman G, Bieri O, Nayak KS. Replication of the bSTAR sequence and open-source implementation. Magn Reson Med. 2024 Apr;91(4):1464-1477.
- bSTAR Pulseq source codes: https://github.com/usc-mrel/replication_bstar
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